

ENGINEERING CHEMISTRY LABORATORY

I Year I Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS10	BSC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100

Course Outcomes:

The experiments will make the student gain skills on:

- Determination of parameters like hardness of water and rate of corrosion of mild steel in various conditions.
- Able to perform methods such as conductometry, potentiometry and pH metry in order to find out the concentrations or equivalence points of acids and bases.
- Students are able to prepare polymers like bakelite and nylon - 6.
- Estimations saponification value, surface tension and viscosity of lubricant oils.

LIST OF EXPERIMENTS	
Experiment-1	Determination of total hardness of water by complexometric method using EDTA
Experiment-2	Determination of Alkalinity of given water sample
Experiment-3	Estimation of Chloride content of water by Argentometry.
Experiment-4	Estimation of the concentration of an acid by Conductometry
Experiment-5	Estimation of amount of ferrous ion by potentiometry using potassium dichromate
Experiment-6	Estimation of HCl by potentiometry
Experiment-7	Preparation of Bakelite
Experiment-8	Preparation of Thiokol Rubber
Experiment-9	Estimation of acid value of given lubricant oil
Experiment-10	Estimation of Viscosity of lubricant oil using Ostwald's Viscometer.
Experiment-11	Determination of rate of corrosion of mild steel in the presence and absence of
Experiment-12	Estimation of Surface tension of a given liquids using Stalagmometer
VIRTUAL LABS	
Experiment-13	Construction of Fuel cell and its working
Experiment-14	Smart materials for Biomedical applications
Experiment-15	Batteries for electrical vehicles
Experiment-16	Functioning of solar cell and its applications
Reference Books:	
1. Lab manual for Engineering chemistry by B. Ramadevi and P. Aparna, S Chand Publications, New Delhi (2022) 2. Vogel's textbook of practical organic chemistry 5th edition 3. Inorganic Quantitative analysis by A.I. Vogel, ELBS Publications. 4. College Practical Chemistry by V.K. Ahluwalia, Narosa Publications Ltd. New Delhi (2007)	